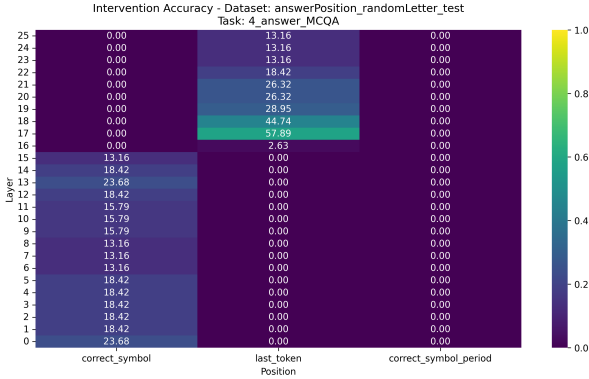


the corresponding token value from the text prompt. For instance, consider the text prompt shown in Figure 7b. First, the X_{Order} variable identifies the position of the correct option—position 3. Then, the O_{Answer} variable uses this information to locate and extract the value of the third option from the prompt—i.e., C —which becomes the final output. Similar high-level causal models have also been proposed in prior work (Lieberum et al., 2023b; Prakash et al., 2024).

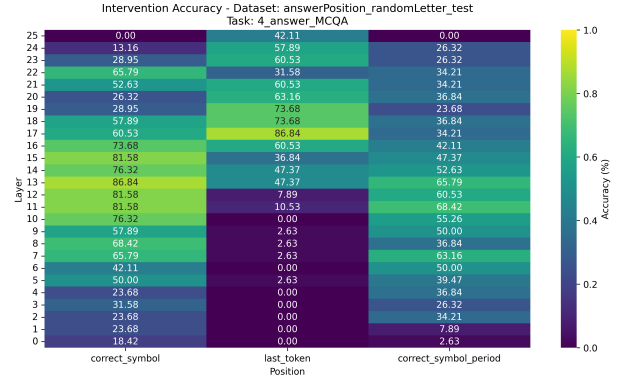
We highlight two interchange intervention experiments—one for each variable in the causal model—to align the LM’s internal representations with those of the causal model. For both variables, we use counterfactuals in which the position of the correct option and the option label is replaced with a different letter. (To be clear, in the main text, we report the average across this counterfactual and counterfactuals that only change position and only change the option label.)

Figures 8a and 8b show the alignment results of the X_{Order} variable in the Gemma model using full vector patching and the DAS method, respectively. This result shows that we can use DAS isolate the answer position in the middle layers above the correct label token and then the final token. This behavior aligns with our hypothesized causal model, in which the model first identifies the position of the correct option, before dereferencing it to fetch token value information.

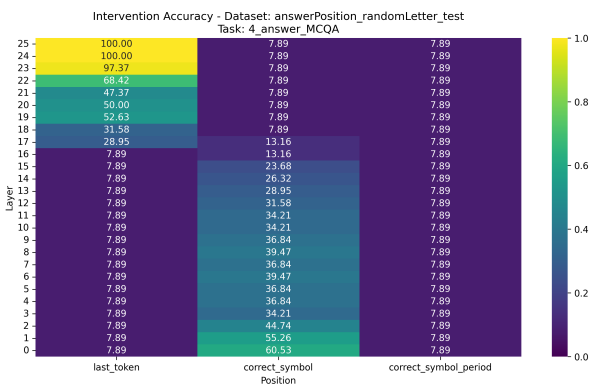
Figures 8c and 8d illustrate the alignment results of the O_{Answer} variable in the Gemma model, using the full vector patching and DAS methods, respectively. The success of full vector patching at the last layer of the last token is logically guaranteed, as this is where the model predicts the next token. However, full vector patching fails at other locations because the output token is tangled with ordering information and the positional encoding. On the other hand, DAS is able to disentangle the output token from this other information at basically every layer.



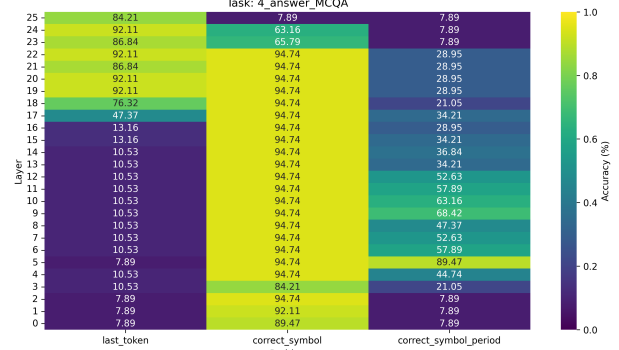
(a) X_{Order} variable alignment results with full residual vector patching.



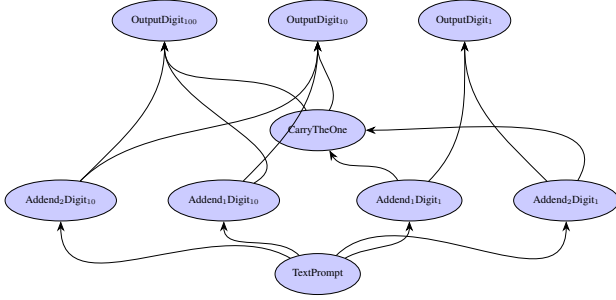
(b) X_{Order} variable alignment results using the subspace identified using the DAS method.



(c) O_{Answer} variable alignment results with full vector patching.



(d) O_{Answer} variable alignment results using the subspace identified using the DAS method.



(a) General causal model for two-digit addition

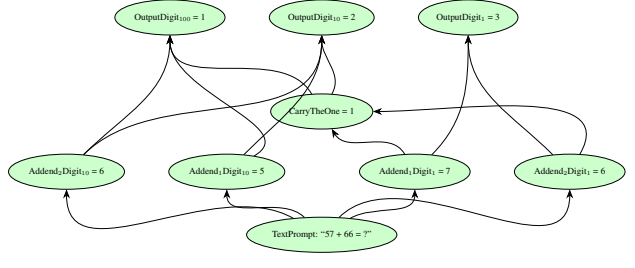
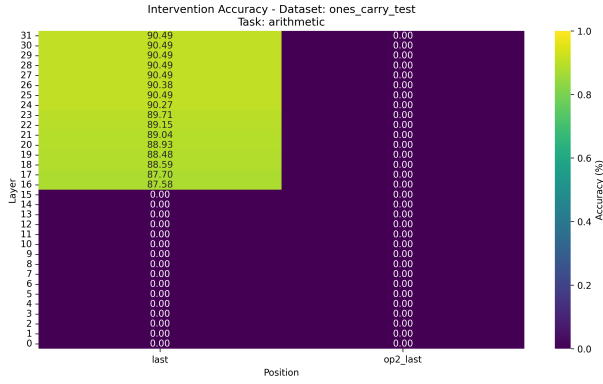
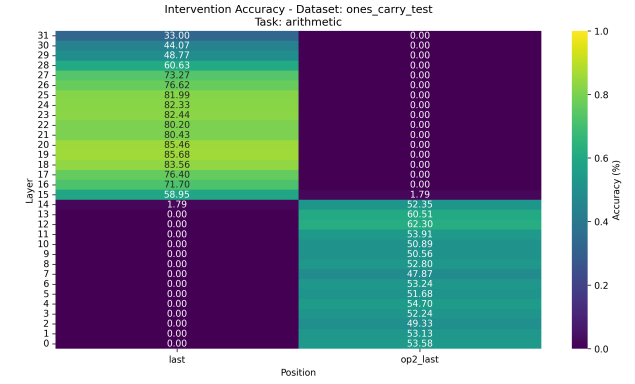
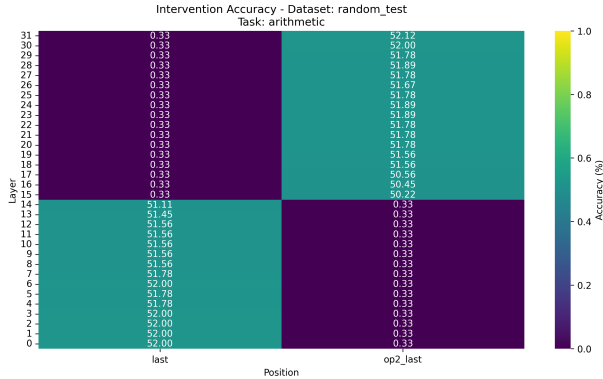
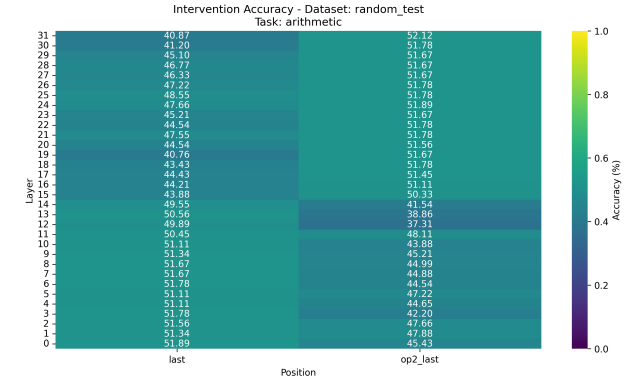

 (b) Specific example: $57 + 66 = 123$

Figure 9. Causal model for two-digit addition arithmetic. The model processes addition through a series of interdependent mechanisms: (1) The 1’s digits from both addends directly influence the output 1’s digit through modular addition. (2) When the sum of 1’s digits exceeds 9, the CarryTheOne variable becomes 1, otherwise it’s 0. (3) The 10’s digit of the result is determined by three inputs: the 10’s digits of both addends and the CarryTheOne value. (4) The 100’s digit is causally influenced by both 10’s digits of the addends and the carry operation—it becomes 1 only when the sum of 10’s digits plus any carried value exceeds 9.


 (a) X_{Carry} variable alignment results in Llama with full vector patching and counterfactuals that differ only by the carry the one variable.

 (b) X_{Carry} variable alignment results in Llama using the DAS method and counterfactuals that differ only by the carry the one variable.

 (c) X_{Carry} variable alignment results in Llama with full vector patching and counterfactuals that are randomly sampled.

 (d) X_{Carry} variable alignment results in Llama with the DAS method and counterfactuals that are randomly sampled.

F.5.2. ARITHMETIC

We define the causal model for arithmetic (addition) in Figure 9. Unlike the causal models for MCQA and ARC, the addition causal model involves multiple variables, as illustrated in Figure 9a. First, the units and tens digits of both addends are parsed. The unit digits are then added together to determine the units digit of the result, as well as whether a carry is

generated. Next, the tens digits of both addends—along with the carry, if any—are summed to compute the tens digit of the result. This step also helps determine whether the result includes a hundred’s digit.

Consider the example shown in Figure 9b, namely $57 + 66$. The causal model begins by parsing the addends to identify their respective tens and units digits. It first adds the unit digits, 7 and 6, determining that the units digit of the result is 3, and that a carry is generated. Next, it adds the tens digits of both addends along with the carry, concluding that the tens digit of the result is 3. Using these same three values—the tens digits and the carry—the model also determines that the result includes a hundreds digit, which is 1.

We evaluate the hypothesized causal model by conducting an interchange intervention experiment to align the X_{Carry} variable. The counterfactual examples are designed such that they introduce a carry when the original does not, and remove it when the original includes one. Figures 10a and 10b present the alignment results in the Llama model using full vector patching and the DAS method, respectively. At first, these results suggest that the X_{Carry} information emerges at the last token position during the middle layers of the model. However, this is likely a representation of the tens-digit output, which perfectly correlates with the ones-carry variable in these counterfactual. Indeed, when we use random counterfactuals in which the output is entirely independent from the carry the one value, we see a total failure.

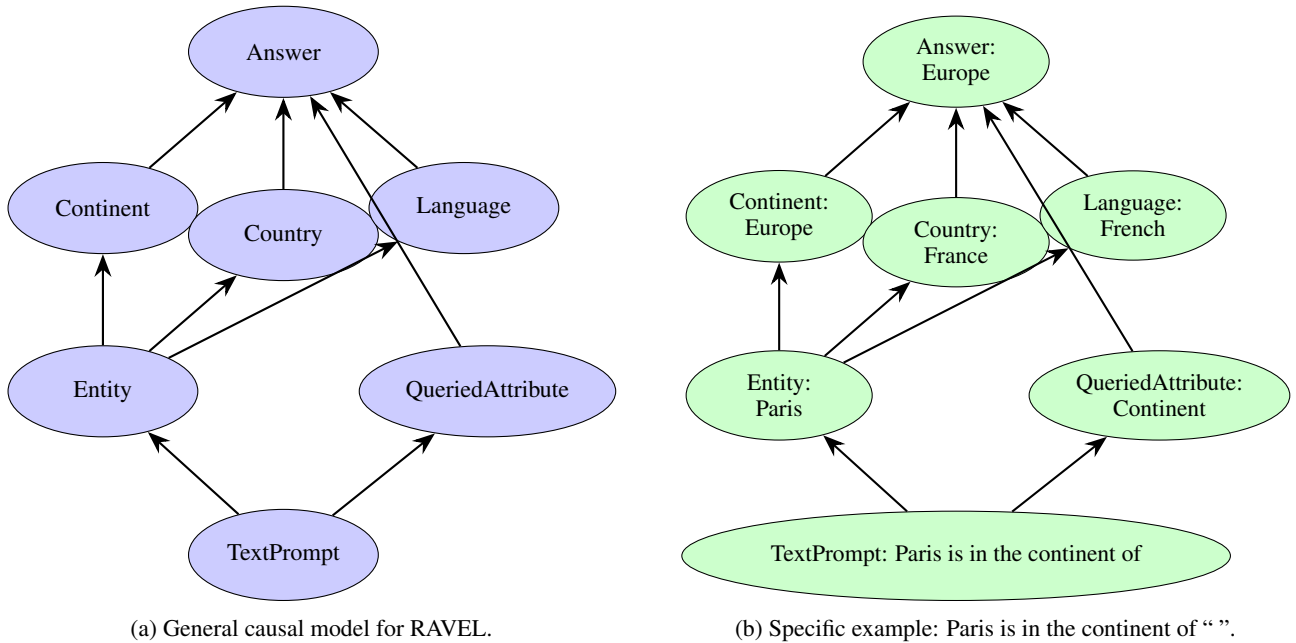


Figure 11. Causal model for the RAVEL task. Given a prompt querying an attribute of a city entity, the model extracts the entity and queried attribute as input variables, then identifies the values of the A_{Cont} , A_{Country} , and A_{Lang} attributes for the entity. Lastly, it decides which value to output based on the queried attribute.

F.5.3. RAVEL

We define the causal model for the RAVEL task in Figure 11. The model takes an input prompt that queries the value of an attribute of an entity (e.g., the continent of a city) and outputs the correct answer. As illustrated in Figure 11a, the causal model first parses the input prompt T to extract two input variables: the *entity* and the *queried attribute* A_{Query} . The entity refers to a city, which is associated with three attribute variables: A_{Cont} , A_{Country} , and A_{Lang} . The model then identifies the value of each attribute variable for the given entity. Lastly, the answer variable O selects the appropriate value based on the queried attribute. For example, given the prompt “Paris is in the continent of” in Figure 11b, the model first identifies the entity “Paris” and the queried attribute “continent.” It then uses its internal knowledge to retrieve the continent, country, and language associated with the city of Paris. Lastly, since the queried attribute is “continent,” the model outputs “Europe.”

The Llama model achieves 66.5% accuracy on the task and the Gemma model achieves 70.5% accuracy on the task. The dataset contains many cities, so considering that there are many countries and languages to choose from, this is good performance. We filter out failure cases.